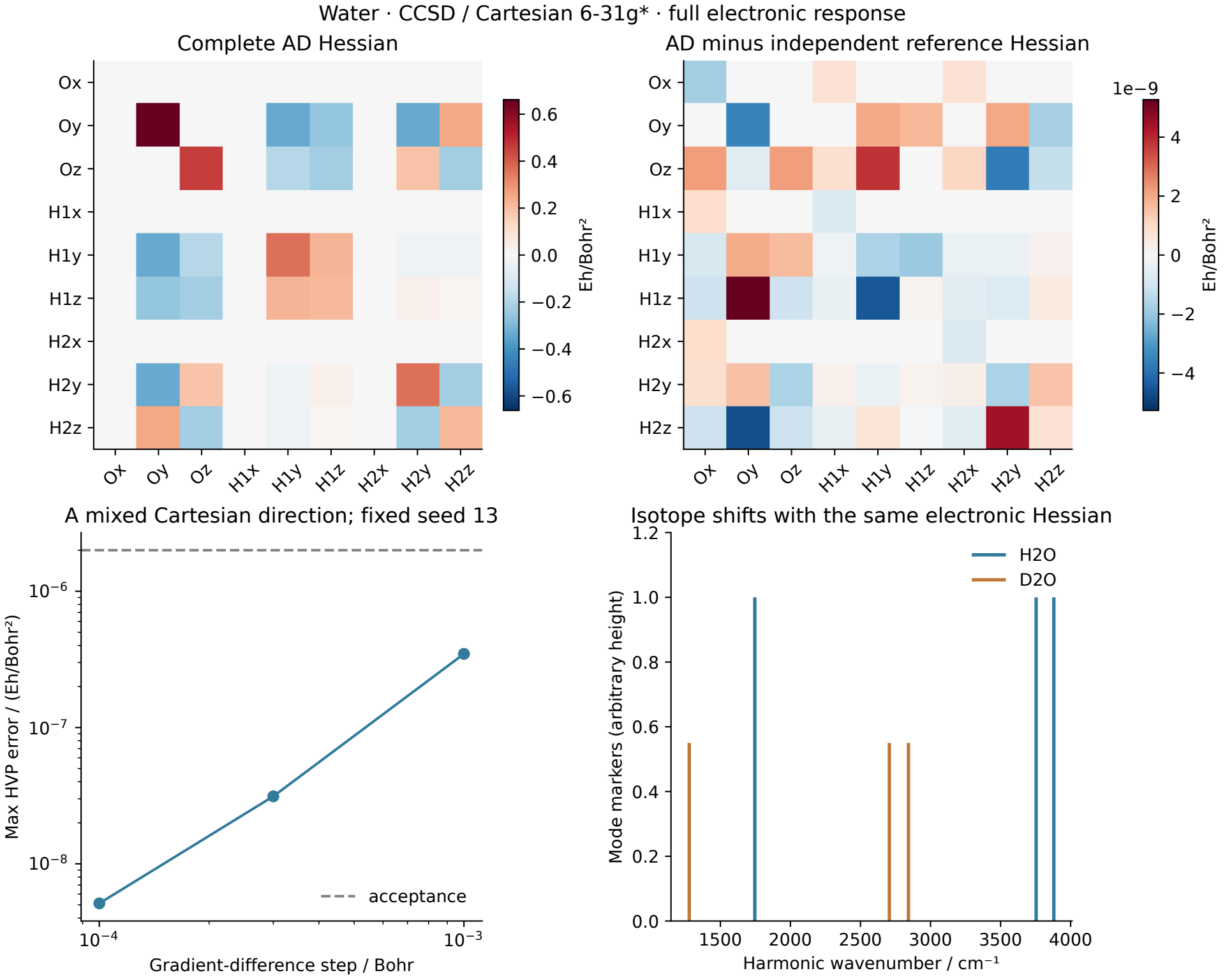
Water · CCSD / Cartesian 6-31g* · full electronic response
Complete AD Hessian
AD minus independent reference Hessian
Ox
Oy
Oz
H1x
H1y
H1z
H2x
H2y
H2z
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Eh/Bohr²
1e−9
4
2
0
−2
−4
A mixed Cartesian direction; fixed seed 13
Max HVP error / (Eh/Bohr²)
10^{-6}
10^{-7}
10^{-8}
acceptance
10^{-4}
10^{-3}
Gradient-difference step / Bohr
Isotope shifts with the same electronic Hessian
H2O
D2O
Mode markers (arbitrary height)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1500
2000
2500
3000
3500
4000
Harmonic wavenumber / cm⁻¹